Case Study 12: Attendance Recovery Computation

1. Problem Statement

Given total conducted classes and attended classes for a student, compute the current attendance percentage and determine the minimum number of additional consecutive classes required to reach a target threshold (e.g., 75%).

2. Objectives

- Calculate real-time attendance percentages accurately.
- Evaluate eligibility against institutionally prescribed attendance thresholds.
- Apply mathematical recovery formulas to determine required future class attendance.

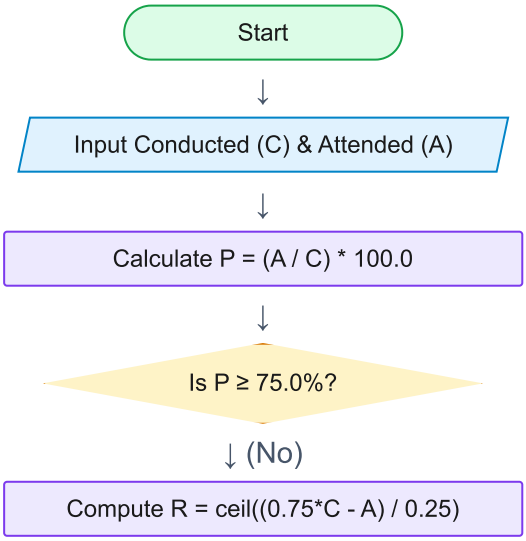
3. Assumptions

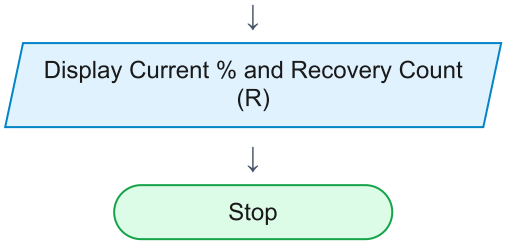
- Minimum target attendance threshold is **75.0%**.
- Total conducted classes `C > 0` and total attended classes `A` satisfies `0 ≤ A ≤ C`.
- Future classes attended incrementally increase both conducted and attended class tallies simultaneously.

4. Algorithm

1. Start.
2. Read total conducted classes `C` and attended classes `A`.
3. Calculate current percentage: `P = (A / C) * 100.0`.
4. If `P >= 75.0`, required additional classes `R = 0`.
5. Else, compute required additional classes: `R = ceil((0.75 * C - A) / 0.25)`.
6. Display current attendance percentage and required recovery classes.
7. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT conducted, attended IF conducted <= 0 OR attended < 0 OR attended > conducted THEN PRINT "Invalid Inputs" EXIT ENDIF current_pct = (attended / conducted) * 100.0 target_pct = 75.0 IF current_pct >= target_pct THEN needed = 0 ELSE // Recovery formula derivation numerator = (0.75 * conducted) - attended needed = CEIL(numerator / 0.25) ENDIF PRINT "Current Attendance: " + current_pct + "%" PRINT "Required Classes to reach 75%: " + needed STOP
```

7. Test Case and Expected Output

Test Case 1: Below Threshold (Conducted = 40, Attended = 25)

```
Current Attendance: 62.50% Target Attendance: 75.0% Status: Below Threshold
Additional Consecutive Classes Required: 5
```

Test Case 2: Above Threshold (Conducted = 50, Attended = 42)

```
Current Attendance: 84.00% Target Attendance: 75.0% Status: Target Achieved
Additional Consecutive Classes Required: 0
```

8. Conclusion

The Attendance Recovery Computation system provides students and academic counselors with actionable insight into minimum attendance targets, facilitating proactive attendance management.